Unit Finger

SKU:U008













Description

Unit Finger is a fingerprint recognition sensor. It integrates the FPC1020A capacitive fingerprint recognition module internally, featuring multiple fingerprint enrollment, image processing, feature extraction, fingerprint matching, and searching functions.

It adopts the UART communication protocol, with a compact appearance design and low power consumption, delivering a reliable security level while providing optimal user convenience for your projects. If you want to add biometric fingerprint recognition to your project with stable and reliable enrollment, verification, and management mechanisms, **Unit Finger** is a good solution.

When in use, please connect this Unit to PORT C, it will communicate with M5Core via the UART protocol.

UART parameter settings:

- Baud rate (**default: 19200bps**)
- Start bit (1 bit)
- Stop bit (1 bit)
- Parity bit (none)

Features

- Uses capacitive area-array semiconductor fingerprint sensor
- Each pixel of the sensor has 256 grayscale levels of pixel quality
- Supports 1:N identification and 1:1 verification functions

- Supports 360-degree finger rotation recognition
- Fingerprint matching and searching functions

Includes

- 1 x Unit Finger
- 1 x HY2.0-4P Grove Cable (20cm)

Applications

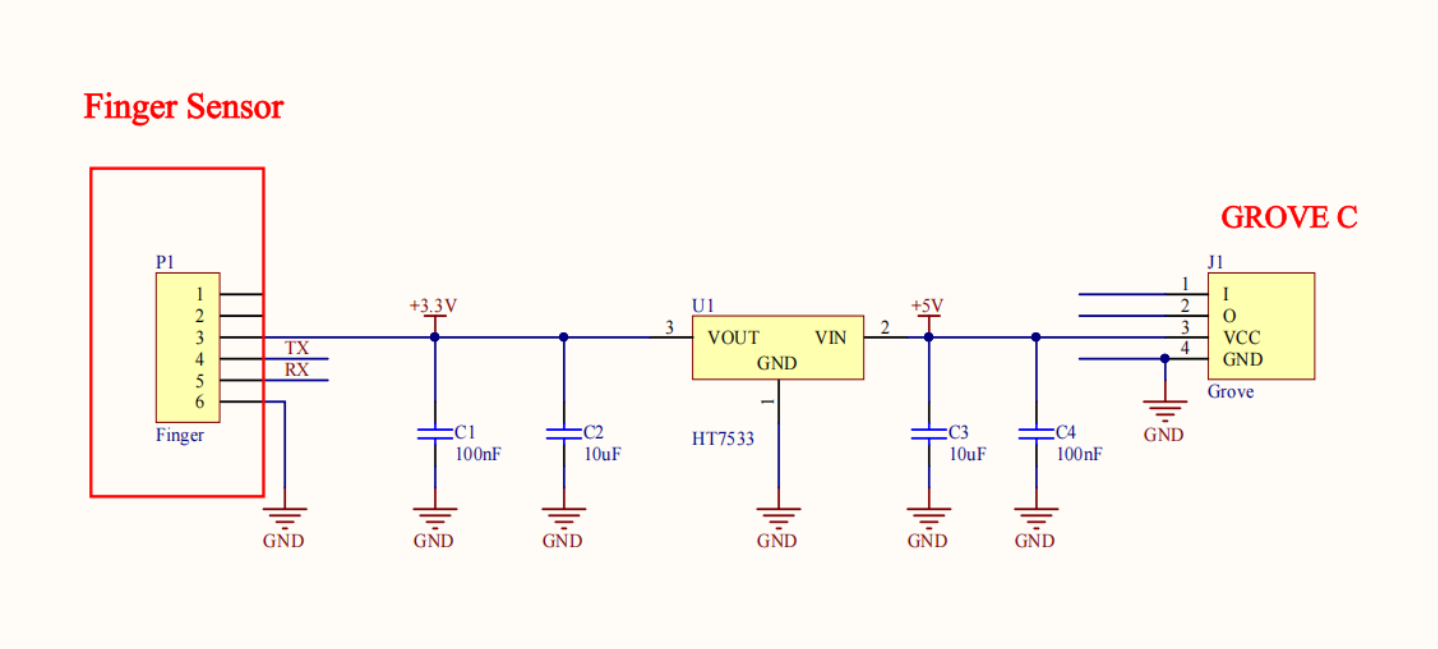
- Fingerprint attendance machine
- Fingerprint storage cabinet

Specifications

Specification	Parameter
Fingerprint Capacity	150
Security Level	0-9, default 5
Output Format	Username, Image, Feature value
Feature Value Size	193Byte
Communication	UART (9600-115200)
Operating Temperature	-10°C ~ 60°C
Relative Humidity	20%-80%
Product Size	48.0 x 24.0 x 8.0mm
Product Weight	7.7g
Package Size	138.0 x 93.0 x 9.0mm
Gross Weight	12.9g

Schematics

- Unit Finger Schematics PDF

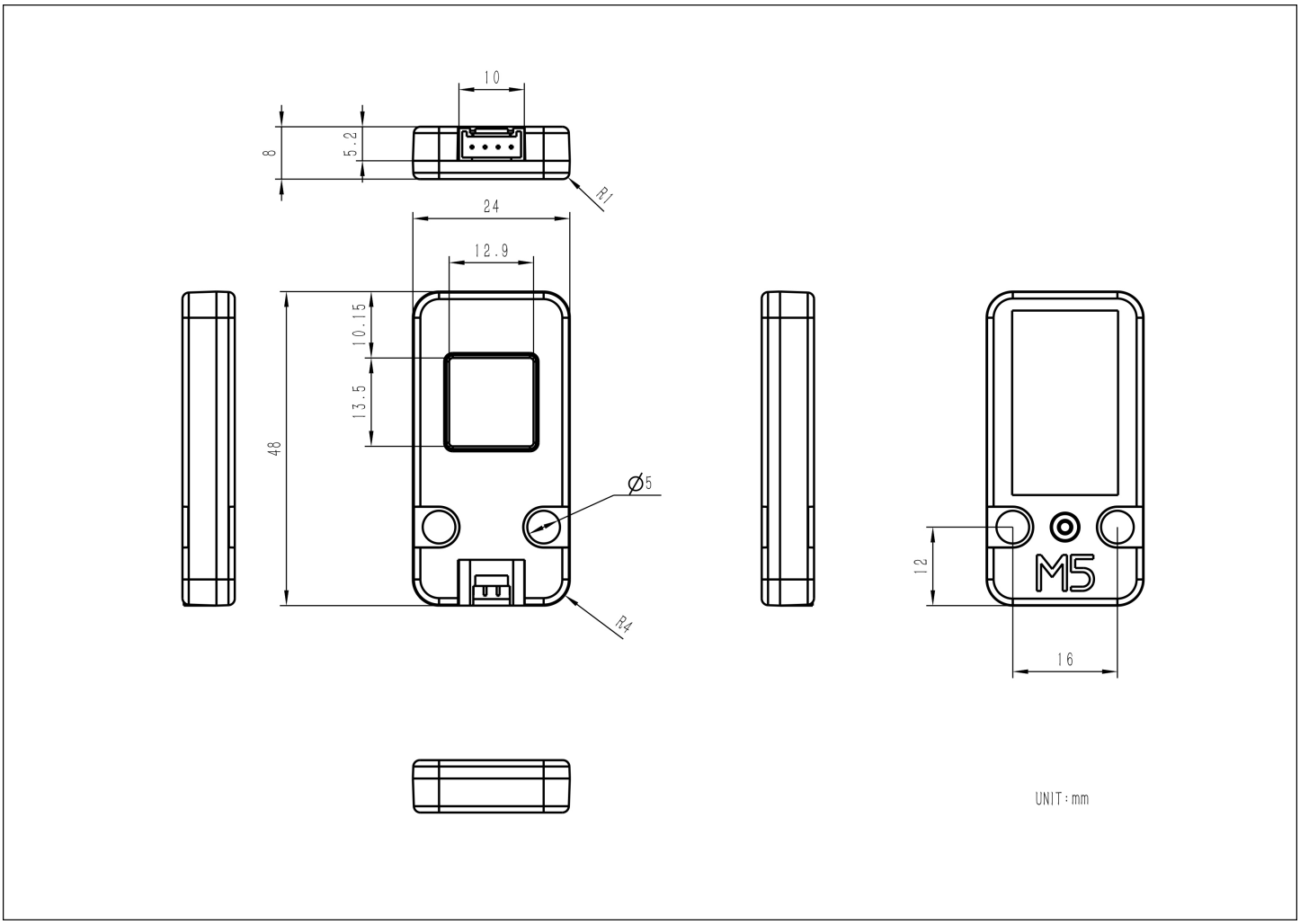


PinMap

Unit Finger

HY2.0-4P	Black	Red	Yellow	White
PORT.C	GND	5V	UART_RX	UART_TX

Model Size



Structure

- Unit Finger Structure Files

Datasheets

- Communication Protocol **FINGER Serial Communication Protocol**
- Datasheet **FPC1020A**

Softwares

Arduino

- Unit Finger Arduino Library
- Unit Finger Arduino Tutorial

UiFlow1

- Unit Finger UiFlow1 Docs

UiFlow2

- Unit Finger UiFlow2 Docs

EasyLoader

Easyloader	Download	Note
Unit Finger Test Easyloader	download	/

Video

- FINGER UNIT usage case: Press the left button to enter fingerprint enrollment mode. Press the middle button to enter fingerprint recognition mode.

Finger_UNIT.mp4

- **FINGER Demonstration**

Fingerprint Unit.mp4

- **UIflow Use**